

A contactless and AI-assisted sensor for gait assessment of stroke patients[☆]

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HIGHLIGHTS

- mmGait: a contactless mmWave sensor tracking 3D foot locations for gait analysis.
- Validated on nine stroke patients and nine healthy controls against OptiTrack.
- Gait-speed error ~0.02 m/s and stride-length error ~3 cm versus motion capture.

ARTICLE INFO

Keywords:

Millimeter-wave
Deep learning
Gait monitoring
Wireless sensing
Foot location prediction

ABSTRACT

The gait parameters, i.e., gait speed, stride length, step length, and cadence, serve as key indicators for patients with mobility impairments. Monitoring these parameters can significantly aid in assessing recovery progress, especially in stroke patients. Compared to traditional clinical visits, wearable sensors, or motion capture systems, millimeter-wave (mmWave) device offers unique advantages: low-cost, contactless, light-free, and no privacy invasion, making frequent gait monitoring feasible for home deployment. We introduce *mmGait*, an mmWave-based system that extracts foot features from reflections and learns walking patterns to accurately predict the 3D locations of both feet, from which we can derive the gait parameters automatically without requiring subjects to wear any sensors or be recorded by cameras. Across multiple stroke patients, *mmGait* achieves low median errors in estimating key gait parameters: less than 0.03 m/s for gait speed, ~3 cm for stride length, 7.4 cm for step length, and below 10 steps/min for cadence. Agreement analysis further demonstrates good agreement with the OptiTrack reference for gait speed and stride length, supporting the feasibility of the proposed contactless approach for coarse spatiotemporal gait assessment.

1. Introduction

Human gait reflects the locomotion of the body's center of gravity and serves as a critical biomarker for overall health status [1]. Key gait parameters, such as speed, step length, stride length, and cadence, provide valuable insights into neuromuscular function and physical well-being [2,3]. In particular, gait speed, often referred to as the "sixth vital sign", is a reliable, valid, and sensitive indicator of health and functional ability [4,5]. Healthy adults typically exhibit a gait speed around 1 m/s, while speeds below 0.6 m/s are linked to increased risks of falls, hospitalization, and mortality. Walking speeds under 0.4 m/s are commonly associated with severe mobility impairments, such as those seen in stroke survivors [2,6]. Importantly, small changes in gait speed, on the

order of 0.1 m/s, can represent clinically meaningful improvements, underscoring the value of continuous gait monitoring in rehabilitation [4]. Step length symmetry, stride length, and cadence are also strongly correlated with motor recovery, providing additional dimensions for assessing rehabilitation progress [7,8].

Conventional gait assessment methods are typically conducted in clinical settings, requiring patients to perform standardized walking tests under professional supervision [2,9]. These methods often involve timed walking trials using a stopwatch and observational assessment by clinicians to evaluate balance and gait quality [10]. However, such approaches are time-consuming, rely on in-person visits, and are difficult to scale for long-term or home-based monitoring. More

[☆] This work was conducted in collaboration with Prisma Health Hospital, and the study was approved by the University of South Carolina Institutional Review Board (Protocol No. Pro00133989; approved February 20, 2024).

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<https://doi.org/10.1016/j.combiomed.2026.111948>

Received 2 June 2026; Received in revised form 8 September 2026; Accepted 16 September 2026

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sophisticated systems like motion capture (MOCAP), including platforms like OptiTrack [11], Vicon [12], Qualisys [13], and Theia [14], offer high-resolution tracking of 3D foot trajectories using infrared cameras and reflective markers [9,15]. While accurate, these systems are expensive and remain restricted to controlled lab environments.

Alternative approaches using commercial off-the-shelf (COTS) sensors, such as RGB and depth cameras, enable vision-based gait tracking but are sensitive to low-light conditions, making them less suitable for around-the-clock home monitoring, especially at night [16,17]. Wearable sensors (e.g., accelerometers, gyroscopes) offer portable solutions for capturing limb movement [16] but present issues of user compliance, discomfort, and forgetfulness, particularly among older adults [18,19]. These limitations have highlighted the need for remote and contactless sensing technologies for unobtrusive, reliable gait monitoring in naturalistic settings.

Radio frequency (RF) based sensing offers a promising alternative. Systems such as WiTrack [20] and WiGait [2] use RF reflections to estimate body movement and gait parameters. WiGait, for instance, tracks walking velocity and stride length using Wi-Fi signals. However, its performance is limited by the low spatial resolution of Wi-Fi, especially for individuals with impaired or slow gait (e.g., gait speeds < 0.4 m/s), and it does not provide precise foot-level tracking required for assessing symmetry and stability.

Millimeter-wave (mmWave) sensing technology addresses many of these limitations. Operating at frequencies between 77–81 GHz, mmWave systems offer centimeter-level range resolution, enabling fine-grained tracking of body movement. Unlike vision-based sensors, mmWave signals are unaffected by lighting conditions and are privacy-preserving, enabling in-home use. Moreover, their compact form factor and low cost make them suitable for deployment outside clinical environments. Prior works have leveraged mmWave for activity and gait-pattern recognition [21,22], multi-user identification [23–25], 3D body mesh reconstruction [26], and coarse gait parameter estimation [27]. However, few existing works focus on accurate foot-level tracking, particularly for individuals with walking disabilities such as stroke patients, where gait patterns are often asymmetric and temporally irregular.

Recent advances, such as mmMesh [26], use biological models like the Skinned Multi-Person Linear (SMPL) [28] representation to reconstruct full-body 3D meshes from sparse mmWave point cloud data (PCD). However, mmMesh struggles to localize foot positions accurately due to limited vertical resolution and specular reflection issues. Additionally, SMPL models are trained on healthy individuals and do not generalize well to stroke patients, whose gait patterns involve asymmetries and irregular movements.

To address this gap, we present *mmGait*, a novel mmWave and deep learning based system designed for accurate and contactless 3D foot localization thereby estimating the gait parameters. Our design involves both ceiling- and ground-mounted device configurations, and the simulation results indicate that ground-level placement significantly improves foot tracking accuracy by reducing vertical motion artifacts [29]. Given the sparse and noisy nature of mmWave reflections, we introduce a deep learning architecture, *mmGaitNet*, that combines dynamic range-angle heatmaps (RAMaps) from the mmWave device (similarly to

a bird's-eye view) with raw frequency domain data to localize foot positions. A multi-head attention mechanism enables the model to learn temporal gait patterns, enhancing height estimation and robustness to signal sparsity.

We evaluate our system using data collected from 9 healthy individuals and 9 stroke patients, with ground-truth foot trajectories obtained using the OptiTrack [11] motion capture system. Experimental results show that, although *mmGait* exhibits moderate absolute errors in foot localization (23.6 cm on average), the predicted trajectories remain temporally consistent with ground-truth motion. This enables the analysis of clinically relevant gait parameters from the estimated foot trajectories. The proposed system achieves low gait-speed estimation errors, together with low median errors for stride length, step length, and cadence. The results indicate that *mmGait* is well suited for contactless monitoring of coarse spatiotemporal gait characteristics in realistic sensing environments. Furthermore, we conduct a preliminary study on stroke-status classification based on gait patterns, which explores the feasibility of distinguishing stroke patients from healthy controls using gait characteristics measured by mmWave sensing.

Our contributions mainly focus on the following aspects: (1) we developed and validated a novel mmWave-based system for contactless foot-level gait analysis, enabling accurate estimation of gait parameters from 3D foot trajectories; (2) we collected a dataset of mmWave gait reflections from stroke patients and controls alongside 3D ground-truth labels; (3) we introduced *mmGaitNet*, a deep learning model that learns temporal walking patterns from mmWave signals to capture inter-foot relationships and support robust gait parameter estimation in realistic environments; and (4) we conducted a feasibility study on distinguishing stroke patients from healthy individuals, demonstrating the feasibility of stroke-status classification using mmWave-derived gait characteristics. Our results demonstrate the feasibility of *mmGait* and offer a promising step toward enabling at-home rehabilitation assessment for individuals with gait disorders. To accelerate contactless research on gait assessment, we will open-source our dataset and codebase.

2. Methods

2.1. Overview

To achieve our goal, we develop *mmGait*, a system that processes raw mmWave signals into dynamic RAMaps (range-angle heatmaps) and FFT data and leverages a deep learning model to predict 3D foot locations for gait analysis. The overall architecture of *mmGait* is illustrated in Fig. 1. To effectively extract features from time-series mmWave data, we design *mmGaitNet*, which takes dynamic RAMaps and FFT data as inputs to capture both global motion cues and detailed signal reflections across time. The dynamic RAMaps highlight moving objects and provide global context, while the FFT data preserve detailed reflections from the environment. To compensate for missing or weak foot reflections in individual frames due to the small reflective surfaces of the feet, our model aggregates 15 consecutive frames as a single sample, enabling it to learn temporal walking patterns, improve consistency across frames, and enhance robustness to sparse and intermittent reflections. Additionally, incorporating multiple frames improves predictions in the

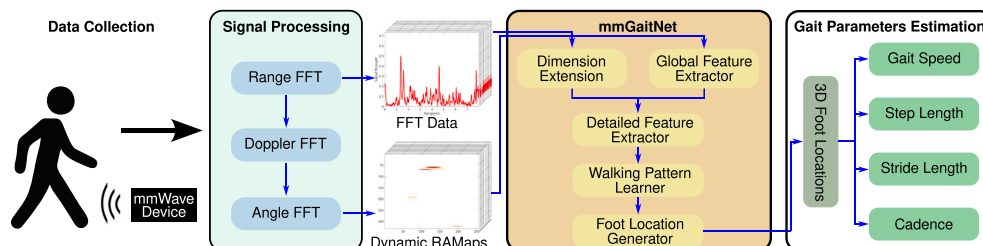


Fig. 1. The overall architecture of *mmGait* system.

vertical dimension, which is inherently limited by the mmWave antenna array.

2.2. Data processing

mmGait utilizes the reflected mmWave signals from human feet to predict the accurate 3D foot locations for monitoring the walking gait of patients. However, the received mmWave reflections, comprising complex time domain signals with amplitude and phase, make it difficult for a deep learning model to directly extract features and learn the walking patterns. Therefore, we first process the raw mmWave signals into FFT data to obtain detailed information of surroundings. Since we are focusing on the walking movement, we further design a data processing algorithm to separate the moving objects from the stationary environment, providing global features to guide the learning process of model. To provide the feedback for optimizing our model, we integrate OptiTrack motion capture system to track 3D foot locations as ground truth. But, the mmWave and OptiTrack data collection programs use different time systems, necessitating a data synchronization program to synchronize the collected dataset.

2.2.1. FFT data and dynamic RAMap generation

As Fig. 1 shows, the data processing part consists of three key steps to generate FFT Data and dynamic RAMaps. *First*, *Range FFT* is applied to the received mmWave reflections from human feet, converting the time domain signals to the frequency domain. The generated FFT data captures detailed distance information across various range bins. *Second*, since the researched target is moving across time-series frames of mmWave data, we apply *Doppler FFT* on multiple frames to distinguish dynamic objects from static surroundings. *Third*, *Angle FFT* is used to extract azimuth angle information from non-overlapping virtual antenna channels of Range-Doppler data, providing the spatial information about foot locations. Specifically, to enhance the quality of the dynamic RAMaps, we apply a thresholding method that retains only the signals with higher strength to remove noise, ensuring more accurate and reliable global information for *mmGaitNet*.

2.2.2. Data synchronization

We use the mmWave device to capture the mmWave reflections and multiple tracking cameras to record the ground-truth foot locations. However, they require different software to capture the data, and starting both programs at the micro-second level is not feasible. To solve this issue, we trigger both programs to start simultaneously and record actual timestamps. We collect mmWave reflections at 50 ms intervals (20 Hz), where one frame refers to one raw mmWave radar frame, and ground-truth locations at the higher OptiTrack frame rate of 120 Hz (see [Section 2.4](#)). This results in a different number of frames for the mmWave data than for the ground truth. To synchronize the data samples, we look at the timestamps of the first data sample from mmWave and ground-truth 3D foot locations from OptiTrack and select the samples starting later as the first synchronized sample. Since mmWave frames are captured at a lower frequency, we find the closest frame from OptiTrack data based on the timestamp and assign it to the mmWave data. We stop when we reach the last frame of either mmWave

or OptiTrack. During synchronization, we ensure that the timestamp difference of each synchronized sample is close to 50 ms and that the time difference between the mmWave frame and the OptiTrack frame is less than 10 ms across all the frames.

2.3. mmGaitNet

Although our mmWave radar provides a high range resolution of 4.9 cm, the low angular resolution in both azimuth and height dimensions [30] makes tracking 3D foot locations using mmWave reflections from human feet. Due to the specularity of mmWave, small reflective surfaces of feet, and limited number of antennas, the insufficient 3D information from foot reflections makes it challenging for our system to predict accurate foot locations for patients with abnormal gait. To address these challenges, we develop an innovative deep learning model *mmGaitNet*, which is designed to extract foot features from processed FFT data and dynamic RAMaps and learn the abnormal walking patterns from multiple time-series frames of mmWave data, ultimately generating accurate 3D locations for both feet. Fig. 1 illustrates the overall architecture of *mmGaitNet*, which consists of five main components: *Global Feature Extractor*, *Dimension Extension*, *Detailed Feature Extractor*, *Walking Pattern Learner*, and *Foot Location Generator*. The *Global Feature Extractor* produces a trainable weight matrix from the dynamic RAMap to guide the *Detailed Feature Extractor* in focusing on valuable foot information. Additionally, the global features generated by this module provide coarse 2D foot locations to facilitate the training process. The extended FFT data from the *Dimension Extension* module, along with the trainable weights, are then passed to the *Detailed Feature Extractor* to capture finer foot features from the mmWave reflections. Next, the *Walking Pattern Learner* processes 15 continuous frames of foot features and global features to learn the abnormal walking patterns of stroke patients. Finally, the learned walking pattern features, combined with global features from *Global Feature Extractor*, are passed to the *Foot Location Generator* module, which outputs the final 3D foot location predictions.

2.3.1. Global feature extractor and dimension extension

As shown in Fig. 2, to increase the feature channels of input, *Global Feature Extractor* first uses a *Focus* block to slice dynamic RAMap into 4 channels with lower resolution and employs *Multilayer Perceptron (MLP)* to expand channels to 32. The *MLP* block in figure represents multiple repeated combinations of convolutional layers, activation layers, and batch normalization layers. By continuously reducing the width and height of feature map, this module extracts global features of human feet, which provide the coarse foot locations for following modules to improve the prediction accuracy and facilitate training process. To preserve the detailed information, we use shortcut connections that concatenate the features from earlier layers for upsampling process. Ultimately, this module produces a trainable weight matrix, which can eliminate or amplify the reflections in FFT data to guide the subsequent learning process. *Dimension Extension* module is designed to extend the dimensions of processed FFT data, ensuring it matches the trainable weights generated by the *Global Feature Extractor* module (Fig. 1).

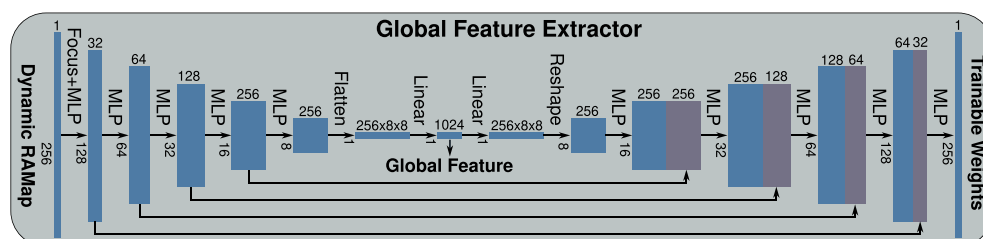


Fig. 2. The structure of global Feature Extractor, which takes dynamic RAMap as input to generate the trainable weight matrix and the global feature.

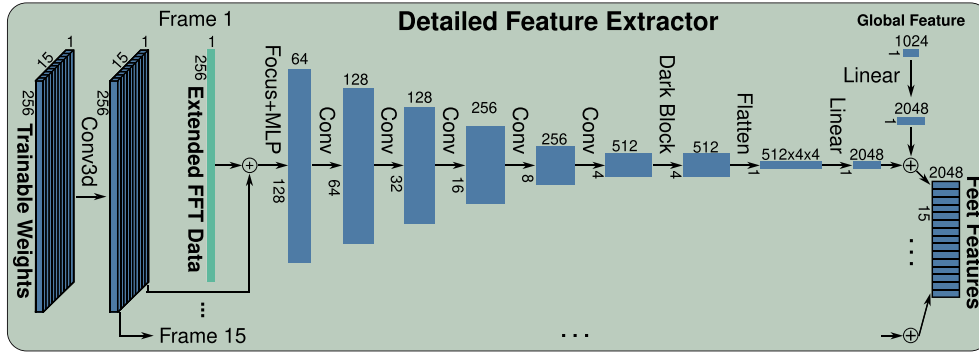


Fig. 3. The architecture of detailed Feature Extractor module.

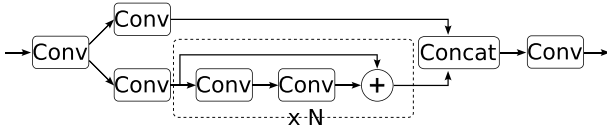


Fig. 4. The detailed structure of Dark Block. We set N to 2 here.

2.3.2. Detailed feature extractor

To extract the accurate foot locations from reflected mmWave signals, *Detailed Feature Extractor* employs multiple *Conv* blocks, consisting of convolution layer, activation layer, and batch normalization layer, to enrich the channels of feature maps (see Fig. 3). The input trainable weight matrix, which is further processed by the *Conv3d* block to combine the time-series information, guides this module to focus on valuable foot features from surroundings. The introduced *Dark Block* [31], following the idea of CSPNet [32], enhances the learning ability of convolutional layers and reduces computational complexity by separating the gradient flow through two different paths. Furthermore, the residual connections, similar to those in ResNet [33], bypass multiple layers to directly add the inputs to outputs, ensuring our deep model learns features effectively. Fig. 4 demonstrates the detailed structure of *Dark Block*. Since we are using 15 frames of mmWave reflections as one sample, we eventually produce 15 frames of foot features in continuous time sequence.

2.3.3. Walking pattern learner

Due to the specularity of mmWave signals and the small reflective surfaces of feet, the reflections from feet may not be received in some frames. And the very limited virtual antenna channels in vertical plane make the 3D prediction task for height difficult. Therefore, we input 15 raw mmWave frames, corresponding to 0.75 s at 20 Hz, as one sample to compensate for the missing information in some frames and learn the foot movement patterns to extract height information. This module *Walking Pattern Learner* is designed to observe the abnormal walking patterns of patients to improve the accuracy of predicted 3D coordinates (see Fig. 5). The core idea of this module is to utilize the *Multi-Head Attention* [34,35] which runs multiple attention mechanisms in parallel to capture intricate patterns from time-series data. To account for the order of frames in a sequence, *Positional Encoding* is added to embed the inputs with absolute positions. Then, the *Feed Forward* module, consisting of fully connected feed-forward layers further extracts foot features.

2.3.4. Foot location generator

Fig. 6 shows the architecture of *Foot Location Generator*. We first employ Linear layers on the stacked 15 frames of global features generated by *Global Feature Extractor* to provide rough locations of feet

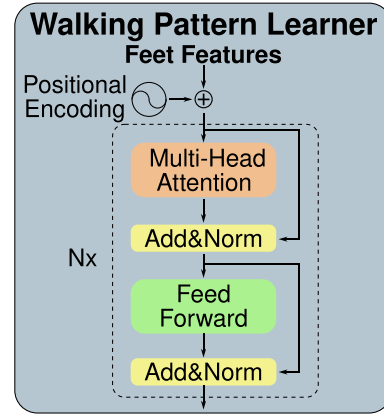


Fig. 5. The architecture of walking Pattern Learner. Specifically, we set N to 4.

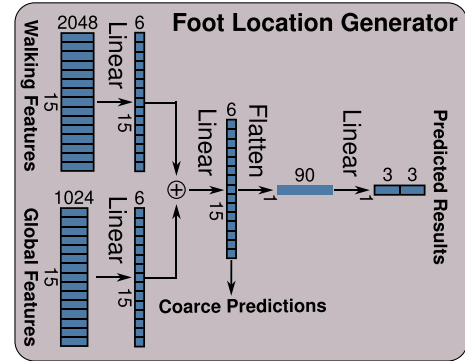


Fig. 6. The architecture of foot Location Generator.

from dynamic RAMaps. Then, we use the extracted walking features to recalibrate the 3D foot locations. The combined features are passed through multiple Linear layers to predict the *Coarse Predictions* for 15 frames. By integrating continuous frames of coarse feet locations, we finally predict the 3D locations of both feet for middle frame. Because a stride-one sliding window is used and each window yields the 3D foot locations of its middle frame, the model produces one prediction per raw frame, so the effective foot-location output rate equals the acquisition rate of 20 Hz.

2.3.5. Loss computation

The 3D locations of the two feet exhibit inherent spatial relationships, as their relative distance is physically constrained. To leverage this

property, we predict the absolute location of one foot and the relative offsets along the x , y , and z coordinates for the other foot. For loss computation, all predictions are first converted to real-world 3D coordinates, and the spatial distance is used to measure the discrepancy between predictions and ground truth.

The Euclidean distance (L2 norm) formulation is defined as:

$$\|I_1 - I_2\|_2 = \sqrt{(I_{1,x} - I_{2,x})^2 + (I_{1,y} - I_{2,y})^2 + (I_{1,z} - I_{2,z})^2} \quad (1)$$

where I_1 and I_2 denote two 3D locations, and $(I_{1,x}, I_{1,y}, I_{1,z})$ and $(I_{2,x}, I_{2,y}, I_{2,z})$ are their corresponding coordinates along the x , y , and z axes.

Since *mmGaitNet* predicts locations for both left and right feet, we define the *Location Loss* as follows:

$$\mathcal{L}_{loc} = \frac{\|\text{loc}_{pre}^l - \text{loc}_{gt}^l\|_2 + \|\text{loc}_{pre}^r - \text{loc}_{gt}^r\|_2}{2} \quad (2)$$

where $\text{loc}_{pre/pt}^{l/r}$ denote the predicted and ground-truth 3D locations of the left and right feet.

To enforce structural consistency between the two feet, we further introduce a *Feet Distance Loss*, which measures the discrepancy between the predicted and ground-truth inter-foot distances:

$$\mathcal{L}_{dis} = \left| \|\text{loc}_{pre}^l - \text{loc}_{pre}^r\|_2 - \|\text{loc}_{gt}^l - \text{loc}_{gt}^r\|_2 \right| \quad (3)$$

To improve the quality of coarse predictions, we incorporate the location supervision over 15 frames using the *Coarse Loss* defined as:

$$\mathcal{L}_{coarse} = \frac{\sum_{i=1}^{15} \mathcal{L}_{loc}^{(i)}}{15} \quad (4)$$

where the i indexes frame.

Finally, the overall loss function is defined as:

$$\mathcal{L} = \alpha \mathcal{L}_{loc} + \beta \mathcal{L}_{dis} + \gamma \mathcal{L}_{coarse} \quad (5)$$

where α , β , and γ are weighting coefficients for the *Location Loss*, *Feet Distance Loss*, and *Coarse Loss*, respectively.

2.4. Hardware

To collect ground-truth 3D coordinates of both feet, we use the OptiTrack [11] motion capture system to track the markers on human feet. Fig. 7(a) shows the deployment of OptiTrack system. This motion capture system is equipped with 16 Prime^X 22 cameras, which offer a 3D location resolution of 0.15 mm and support high frame-rate capture at 360 FPS, along with 2 Prime Color cameras to record RGB video for reference. To distinguish the two feet, we placed 12.7 mm-diameter reflective markers on the upper-front surface of each foot, with three markers on the left foot and five markers on the right foot. The predicted foot position in this study is defined as the geometric center of

the OptiTrack marker cluster for each foot, which provides a consistent marker-based foot location. The OptiTrack system is configured with a frame rate of 120 Hz to continuously capture the foot locations during data collection. Since the time interval between frames is not constant, we use the recorded timestamps for data synchronization.

To collect mmWave reflections from human feet, we use the 77–81 GHz TI IWR1443BOOST [36] positioned approximately 0.4 m above the ground. Fig. 7(a) and (b) demonstrate the mounting position and layout of mmWave radar. The IWR1443BOOST has 4 receiving antennas and 3 transmitting antennas, generating 12 virtual channels. The FMCW parameters of radar are as follows: ramp start frequency, 77 GHz; frequency slope, 29.982 MHz/ μ s; number of ADC samples, 256; sampling rate, 2.5 MHz/s; chirp duration, 230 μ s; frame rate, 20 Hz; and maximum receive antenna gain, 30 dB. Our mmWave device is capable of capturing reflections from target over a range of 12.4 m, providing a range resolution of 4.9 cm.

2.5. Data collection

We collected data from 9 stroke patients with varying levels of impairment and 9 healthy controls. Stroke participants were recruited from the Neurology Stroke Clinic at the Prisma Health Comprehensive Stroke Center. Potentially eligible patients were screened by the study coordinator according to the predefined eligibility criteria, with eligibility confirmed by the physical therapist. Eligible patients were informed about the study, and written informed consent was obtained before enrollment and gait assessment. Healthy-control participants were recruited through the University of South Carolina student and staff mailing lists and provided written informed consent before participation. Tables 1 and 2 summarize the demographic and clinical characteristics of the stroke patients and healthy controls, respectively. The stroke patients had a mean age of 62.6 years, compared with 28.4 years for the healthy controls. Stroke severity was assessed using the National Institutes of Health Stroke Scale (NIHSS), with lower scores indicating less severe neurological impairment. The patient group had a median NIHSS score of 3 (range: 2–7), whereas all healthy controls had a score of 0. Motor function was further characterized using the Fugl-Meyer Assessment Upper Extremity (FMA-UE) and Fugl-Meyer Assessment Lower Extremity (FMA-LE), with maximum scores of 26 and 28 per side, respectively. Higher FMA scores indicate better motor function, and the scores were reported separately for the left and right sides. The stroke cohort also covered a broad range of time post-stroke, from 3 to 150 months.

The experiment was conducted in a setting that simulates real home environments, featuring various objects such as a queen-sized bed, table, chair, and tripods (see Fig. 7[a]). The mmWave transceiver, mounted on a supporting base, was positioned in the corner of OptiTrack system. Doctors from the hospital supervised the walking experiments, instructing each subject to complete at least two trials of the 5 m walking test.

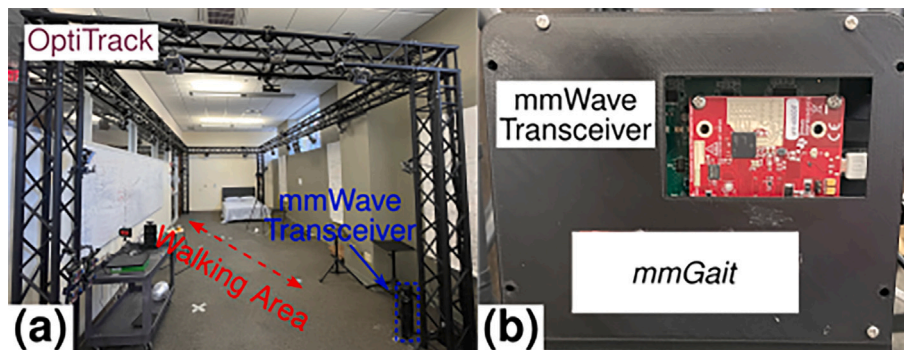


Fig. 7. (a) the experimental setup with OptiTrack system and mmWave radar for data collection. (b) mmWave transceiver inside a protective box.

Table 1

Demographic and clinical characteristics of the stroke patients. C: Caucasian; AA: African American. The FMA-UE and FMA-LE columns report the left-side and right-side scores as left/right.

Cases	Age (years)	Race	Sex	Time post-stroke	NIHSS	FMA-UE (L/R)	FMA-LE (L/R)
P1	65	AA	F	45 mo	4	15/25	16/24
P2	55	AA	F	39 mo	3	26/0	26/25
P3	58	AA	F	26 mo	2	26/5	26/18
P4	59	AA	F	34 mo	4	24/1	26/22
P5	81	C	F	55 mo	3	23/22	24/13
P6	40	C	F	8 mo	2	26/26	28/27
P7	57	AA	F	18 mo	2	26/2	24/19
P8	72	AA	F	150 mo	4	22/14	24/19
P9	76	C	F	3 mo	7	15/23	19/24

Table 2

Demographic and clinical characteristics of the healthy controls. C: Caucasian; AA: African American; A: Asian; H: Hispanic. The FMA-UE and FMA-LE columns report the left-side and right-side scores as left/right.

Cases	Age (years)	Race	Sex	NIHSS	FMA-UE (L/R)	FMA-LE (L/R)
V1	21	A	M	0	26/26	28/28
V2	27	C	F	0	26/26	28/28
V3	29	C	F	0	26/26	28/28
V4	69	AA	F	0	26/26	28/28
V5	24	AA	F	0	26/26	28/28
V6	22	C/H	M	0	26/26	28/28
V7	19	C	F	0	26/26	28/28
V8	21	C	F	0	26/26	28/28
V9	24	C	F	0	26/26	28/28

In each trial, the subject walked from the starting point to the ending point toward the mmWave device and then walked back to the starting point, moving away from the mmWave device. Subjects were instructed to walk at their usual comfortable speed, and no assistive devices were used during data collection. To preserve natural gait behavior and increase usable walking data, we retained the walking-forward, walking-backward, turning, acceleration, and deceleration phases in the dataset. Although reflective markers may contribute weak mmWave reflections, their small size relative to the feet and lower limbs suggests that the measured mmWave signal was dominated by reflections from the participant rather than the markers.

During data collection, all mmWave radar frames were available. Foot markers were occasionally occluded from the OptiTrack cameras, particularly when participants walked near the boundary of the capture volume. For short gaps in the OptiTrack foot-marker trajectories, we applied linear interpolation to estimate the missing 3D foot locations. Temporal segments containing longer gaps or unreliable OptiTrack measurements that could not be appropriately interpolated were removed to ensure high-quality ground-truth foot locations. Table 3 summarizes the number and percentage of interpolated OptiTrack foot-location labels and the number of valid processed samples for each subject. After synchronization, the continuous mmWave and OptiTrack data were segmented into overlapping temporal windows consisting of 15 consecutive frames with a stride of one frame (14-frame overlap). Each processed sample therefore corresponds to one temporal window rather than a single frame. Because consecutive windows overlap by 14 frames, these processed samples are not statistically independent observations, and the effective amount of independent walking data per subject is correspondingly smaller than the raw sample count.

In total, we collected 10,204 processed samples from patients and controls. Table 3 summarizes the number of valid processed samples for each subject after OptiTrack-label interpolation and quality control. Specifically, control subject 6 had no valid processed samples due to the improper OptiTrack marker attachment. Because the patients had

Table 3

Summary of interpolated OptiTrack foot-location labels and valid processed samples for each participant.

Cases	P1	P2	P3	P4	P5	P6	P7	P8	P9
# of Interpolated OptiTrack Frames	24	4	36	1	19	1	1	1	9
Interpolated OptiTrack Frames (%)	6.0	0.3	2.4	0.1	2.1	0.2	0.2	0.5	0.7
# of Processed Samples	356	1258	1500	922	878	546	527	195	1159
Cases	V1	V2	V3	V4	V5	V6	V7	V8	V9
# of Interpolated OptiTrack Frames	33	146	17	0	14	–	84	2	5
Interpolated OptiTrack Frames (%)	11.6	34.1	4.2	0.0	5.2	–	15.1	0.6	1.7
# of Processed Samples	243	400	381	445	240	0	542	346	266

limited ability to go through multiple trials for data collection, we first collected data from 8 volunteers simulating various walking styles to pretrain our model to capture diverse walking patterns. This additional dataset includes 109,144 samples, which were allocated 85,899 for training and 23,245 for testing to learn walking patterns. Subsequently, the pretrained model was fine-tuned separately on the stroke-patient and healthy-control datasets. Because the number of processed samples varied substantially across subjects, we adopted a leave-one-subject-out cross-validation protocol within each dataset to comprehensively evaluate the model's performance. For each leave-one-subject-out fold, an independent copy of the same pretrained model was fine-tuned using all processed samples from the remaining subjects in the corresponding dataset. All samples from the held-out subject were excluded from fine-tuning and used only for validation. The fine-tuned weights from one fold were not carried forward to any subsequent fold; thus, every fold started from the same pretrained initialization and was evaluated independently.

All experimental procedures were approved by the University of South Carolina Institutional Review Board (Protocol No. Pro00133989), and informed consent was obtained from each participant prior to data collection.

To process the mmWave data, FFT-based preprocessing and dynamic RAMap generation were implemented using MATLAB R2024b and conducted on a workstation equipped with an Intel Xeon E5-1620 v4 CPU (3.5 GHz) and 32 GB of RAM. To train and fine-tune the proposed mmGaitNet model, we used a separate GPU server equipped with an AMD EPYC 7282 CPU and an NVIDIA RTX A6000 GPU with 48 GB of memory. The codebase was implemented using Python 3.8.8 and PyTorch 1.12.1 on an Ubuntu 20.04.6 LTS system with a CUDA 11.3 environment.

3. Results

3.1. Evaluation summary

To evaluate the generalization and robustness of our model, we fine-tuned the models separately for the patient and control groups using repeated subject-level validation. In each repetition, one subject from the corresponding group was selected as the validation subject, and all processed samples from this subject were excluded from training and used only for validation. Across repetitions, all subjects in both groups were evaluated as held-out validation subjects. The model achieves an average distance error of 23.6 cm across all subjects, with 19.7 cm for stroke patients and higher error of 27.9 cm for healthy controls primarily due to faster walking speeds and reduced temporal continuity. Despite moderate localization error, the predicted trajectories remain temporally consistent with ground-truth motion, enabling accurate recovery of gait dynamics. Consequently, mmGait achieves low median errors in gait parameter estimation, including 0.022 m/s for gait speed, 3.1 cm for stride length, 7.4 cm for step length, and a median cadence error of 9 steps/min for patients and near-zero for controls. Agreement analysis

Table 4

Average distance error (ADE) and average height error (AHE) of stroke patients and healthy controls.

Group	ADE (cm, mean $\pm$ SD)	AHE (cm, mean $\pm$ SD)
Patients	19.7 $\pm$ 2.9	1.5 $\pm$ 0.3
Controls	27.9 $\pm$ 6.0	2.4 $\pm$ 0.9

Table 5

Average distance error (ADE) and average gait speed (AGS) for stroke patients and healthy controls.

Cases	P1	P2	P3	P4	P5	P6	P7	P8	P9
ADE (cm)	18.1	18.9	22.3	19.3	25.2	20.5	19.0	14.5	20.0
AGS (m/s)	0.17	0.50	0.23	0.57	0.31	0.72	0.68	0.51	0.85
Cases	V1	V2	V3	V4	V5	V6	V7	V8	V9
ADE (cm)	41.6	24.6	24.7	26.5	26.4	–	26.4	22.3	30.3
AGS (m/s)	1.35	0.95	1.25	0.94	1.28	–	1.09	0.96	1.25

further demonstrates excellent agreement for gait speed and good agreement for stride length, whereas cadence and particularly step length exhibit lower agreement with the OptiTrack reference. Overall, these results indicate that mmGait effectively captures the coarse spatiotemporal characteristics of gait and supports reliable estimation of clinically relevant gait parameters, especially gait speed and stride length.

3.2. Error in predicting 3D foot locations

The primary objective of mmGait is to accurately predict the 3D locations of both feet to enable continuous gait monitoring. We evaluate localization performance using the Euclidean distance between predicted and OptiTrack ground-truth foot positions. As shown in Table 4, our model achieves an average distance error of 19.7 ± 2.9 cm for stroke patients and 27.9 ± 6.0 cm for healthy controls. An analysis of the error components shows that the range dimension contributes the largest portion of the localization error, which is expected as range estimation is more sensitive to noise, multipath effects, and to peak localization uncertainty in mmWave signals.

The higher error observed in healthy controls can be partially attributed to their faster walking speeds. As shown in Table 5, the control group exhibited higher gait speeds, ranging from 0.94 to 1.35 m/s, compared with the stroke-patient group, whose gait speeds ranged from 0.17 to 0.85 m/s. Because both groups were recorded at the same frame rate, faster walking results in greater foot displacement between consecutive frames, making temporal tracking of foot locations more challenging. Specifically, V1 had the highest gait speed (1.35 m/s) and the largest localization error (41.6 cm). In contrast, the slower gait speeds of stroke patients allow the model to better leverage temporal information for foot-location estimation.

The reported localization errors aggregate both walking directions, since each trial contains a walking-toward and a walking-away phase.

The two directions present different reflection geometries: when a participant walks toward the device, the dorsal foot surface faces the radar and produces stronger, more direct reflections, whereas walking away exposes the heel and a smaller effective reflecting surface, which can reduce foot visibility. Because mmGaitNet is trained on temporal windows drawn from both directions, it learns a largely direction-agnostic foot representation, and the temporally consistent inter-foot trajectories in Fig. 8(a) and (c) indicate that both phases are tracked without systematic gaps. A formal per-direction error breakdown was not conducted in this study and is left to future work with a larger dataset. Although the absolute localization error is higher than that of vision-based motion capture systems, the high spatial precision is not necessary for gait analysis tasks. The estimation of gait metrics, such as step length, stride length, and gait symmetry, primarily depends on relative motion patterns rather than precise absolute coordinates. Therefore, consistent tracking of foot trajectories is more critical than exact spatial accuracy, as further validated by the low estimation errors in gait parameters (Section 3.4).

A key challenge in mmWave-based sensing is the limited vertical resolution, as the device provides only sparse information along the height dimension. Despite this limitation, mmGait achieves a low average height error of 1.5 cm for stroke patients and 2.4 cm for healthy controls, as shown in Table 4. This result is particularly significant given the inherent sensing limitations, demonstrating that the proposed temporal modeling approach effectively reconstructs height information from sparse measurements by leveraging motion patterns over time. Overall, these results demonstrate that mmGaitNet achieves reliable 3D foot localization under realistic sensing constraints, providing sufficiently accurate and temporally consistent inputs for robust gait parameter estimation.

3.3. Performance on walking patterns

Human walking is a coordinated process involving the central nervous system, bones, and muscles of the lower limbs [37]. The gait cycle consists of alternating stance and swing phases, during which the relative positions of the feet evolve continuously [38]. Consequently, the temporal variation of inter-foot distance provides a compact representation of gait dynamics, capturing key characteristics such as symmetry and stability [39,40].

To evaluate whether mmGaitNet captures meaningful temporal gait dynamics, we analyze the variation of inter-foot distance over time. At each frame, the inter-foot distance is calculated as the Euclidean distance between the 3D locations of the left and right feet, incorporating the differences along all three spatial dimensions. The same calculation is applied to the predicted foot locations and the OptiTrack ground-truth locations. This quantity is used to visualize the temporal variation of the physical separation between the two feet. Fig. 8(a) and (b) show examples from stroke patients P2 and P4, where the ground-truth inter-foot distance exhibits irregular and uneven peak patterns, indicating asymmetrical and unstable gait. The predicted inter-foot distance trajectories show close correspondence with the ground-truth variations, with Pearson correlation coefficients of 0.781 for P2 and 0.776 for P4 and root mean square errors (RMSEs) of 5.6 and 4.3 cm, respectively. These results quantify the similarity of the predicted traces to

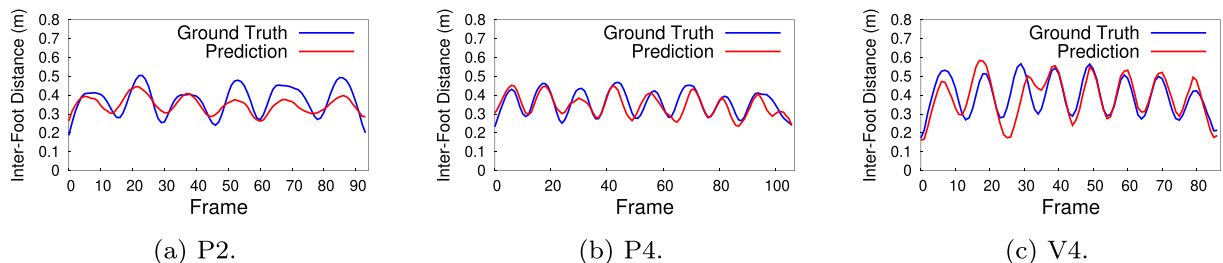


Fig. 8. Examples of inter-foot distance changes over time from (a) P2, (b) P4, and (c) V4.

the OptiTrack reference in both their temporal variations and absolute distances. This demonstrates that the model effectively captures pathological gait characteristics associated with impaired motor control.

In contrast, Fig. 8(c) shows a healthy control subject (V4), where the inter-foot distance exhibits a more periodic pattern corresponding to a stable and symmetrical gait cycle. Notably, the ground-truth curve shows larger peak amplitudes, reflecting longer stride lengths and natural variability in healthy walking dynamics. The predicted curve captures the overall periodicity and amplitude variations, with a Pearson correlation coefficient of 0.727 and an RMSE of 7.9 cm relative to the OptiTrack reference. *These results indicate that mmGaitNet captures not only spatial foot positions but also the underlying temporal structure and variability of gait. Importantly, the ability to track both pathological asymmetry and natural variations in gait is critical for reliable assessment in real-world scenarios.*

3.4. Performance on gait parameter estimation

To evaluate whether the predicted foot trajectories support gait analysis, we derived four gait parameters from the predicted 3D foot locations: gait speed, step length, stride length, and cadence. The same computation procedure was applied to the OptiTrack ground-truth foot trajectories to obtain the reference gait parameters for comparison. We first identified the effective walking period by evaluating the absolute displacement within a short sliding window, thereby separating walking segments from standing or other non-walking periods. Gait speed was then computed as the walking distance during the detected walking period divided by the corresponding walking time. For the remaining gait parameters, we computed the temporal inter-foot distance between the predicted left and right foot positions throughout the detected walking period. The inter-foot distance sequence was smoothed using a moving-average filter, and local peaks were detected and treated as step events. In this study, step length is defined as the Euclidean distance between the left and right foot positions at each detected inter-foot-distance peak. Stride length was computed as the displacement of the same foot between two consecutive detected peak events. Stride length and step length were calculated separately for the left and right feet. Cadence was calculated by counting the detected step events within the walking period and converting the result to steps per minute.

The gait speeds of stroke patients range from 0 to 0.9 m/s, reflecting varying levels of functional impairment, while healthy individuals typically exhibit higher walking speeds exceeding 0.9 m/s (see Table 5). Fig. 9(a) and (b) present the gait speed estimation results for patients and controls, respectively. The scatter plots show a strong correlation between predicted and ground-truth gait speeds, with most points closely aligned along the diagonal line. The Pearson correlation coefficient (PCC) reaches 0.992 for patients and 0.934 for healthy controls, indicating that the model accurately preserves relative differences in walking speed across individuals.

Across all held-out walking trials, the pooled median absolute gait-speed error was 0.021 m/s for stroke patients and 0.038 m/s for healthy

Table 6

Distribution of subject-level median absolute errors, summarized by the 25th, 50th, and 75th percentiles within each group.

Group	Parameter	25th	50th	75th
Patients ($n = 9$)	Gait speed (m/s)	0.014	0.017	0.026
Controls ($n = 8$)	Gait speed (m/s)	0.025	0.046	0.081
Patients ($n = 9$)	Stride length (cm)	1.6	3.0	4.0
Controls ($n = 8$)	Stride length (cm)	1.8	3.0	4.0
Patients ($n = 9$)	Step length (cm)	3.1	6.2	8.9
Controls ($n = 8$)	Step length (cm)	6.3	10.8	12.9
Patients ($n = 9$)	Cadence (steps/min)	0	8.63	12.77
Controls ($n = 8$)	Cadence (steps/min)	0	0	0

controls. To separately characterize between-subject variability, we calculated the median absolute gait-speed error for each participant and then summarized these subject-level median errors using their percentiles. For stroke patients, the 25th, 50th, and 75th percentiles were 0.014, 0.017, and 0.026 m/s, respectively. The corresponding values for healthy controls were 0.025, 0.046, and 0.081 m/s (see Table 6). Given that changes in gait speed on the order of 0.1 m/s are considered clinically meaningful, these results suggest that mmGait can reliably detect subtle changes in gait speed that are indicative of rehabilitation progress. To evaluate absolute agreement with the OptiTrack reference, we further calculated the intraclass correlation coefficient (ICC) using a two-way random-effects model for absolute agreement of single measurements, with 95% confidence intervals. For gait speed, Table 9 reports RMSE values of 0.029 m/s for patients and 0.071 m/s for controls, with corresponding ICC values of 0.992 and 0.931, respectively. These results demonstrate excellent agreement between the gait speeds estimated by mmGait and those derived from the OptiTrack reference system.

Fig. 10(a) and (b) show the distributions of estimation errors across all held-out walking trials for stroke patients and healthy controls, respectively. The stride length error achieves low median values of 3.2 cm for patients and 3.0 cm for controls, indicating that the model effectively captures the spatial consistency of foot placement. The step-length estimation error remains low, with median values of 7.0 cm for patients and 10.9 cm for controls, reflecting reasonable estimation of step-to-step movement. Although occasional step miscounts occur in a small number of patient cases, the overall cadence estimation remains accurate, with a median error of 9 steps/min, while most control cases achieve zero error. The participant-level percentile summaries in Table 6 are consistent with these trial-level median results and quantify the between-subject variability for stride length, step length, and cadence. Because cadence is obtained directly from the detected step events, the cadence agreement reported in Table 9 also serves as an indirect validation of step-event detection against the OptiTrack reference: the high control cadence agreement (ICC = 0.860) confirms that step events are reliably detected for periodic gait, whereas the lower patient agreement (ICC = 0.469) reflects the occasional miscounts that arise under irregular,

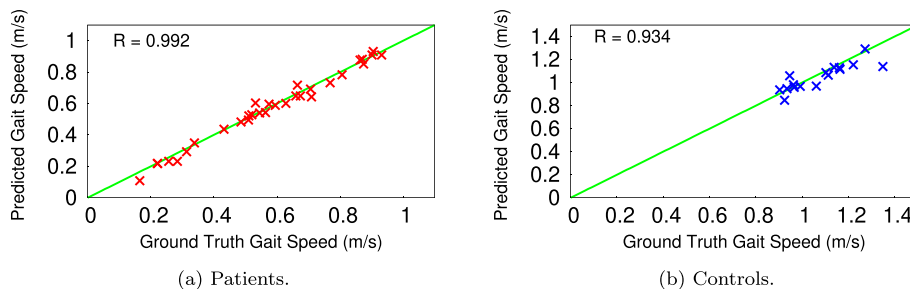


Fig. 9. The scatter plots of gait speed estimation for (a) patients and (b) controls. The green line shows where the predicted walking speeds exactly match the ground-truth values. The R value denotes the PCC.

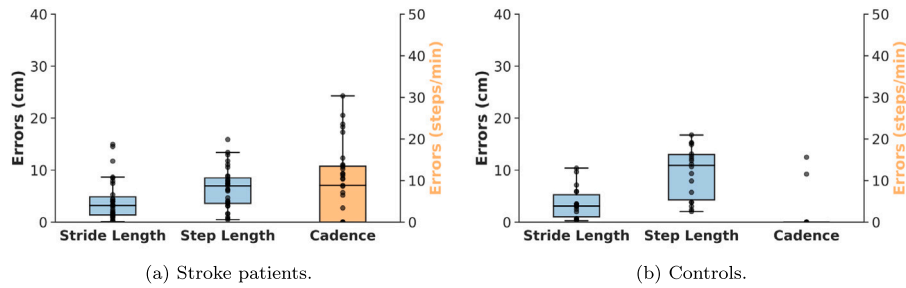


Fig. 10. The error box plots of stride length, step length, and cadence across all walking tests for (a) stroke patients and (b) controls. Specifically, the unit of cadence is steps/min corresponding to the right y-axis.

Table 7

Examples of calculated stride length, step length, and cadence for stroke patients and a healthy control. The predictions and ground truths for P2, P4, and V4 are illustrated for comparison. Specifically, all values are averaged across all steps of experimental test.

Cases	P2	P4	V4
True Gait Speed (m/s)	0.496	0.570	0.939
Predicted (m/s)	0.514	0.567	0.932
True Stride Length (cm)	40.3	40.1	52.9
Predicted (cm)	35.9	36.8	54.7
True Step Length (cm)	42.8	43.3	52.0
Predicted (cm)	37.0	40.4	49.9
True Cadence (steps/min)	78	88	107
Predicted (steps/min)	88	92	104

Table 8

Examples of P2, P4, and V4 to display the stride length changes over steps.

Steps	1	2	3	4	5
Stride Length of P2 (cm)	32.1	45.9	29.5	48.9	30.9
Stride Length of P4 (cm)	30.9	47.1	32.3	66.1	39.1
Stride Length of V4 (cm)	49.7	35.9	80.6	96.4	74.0

asymmetric gait. A step-by-step temporal-alignment validation of individual gait events would require higher-rate acquisition and is beyond the scope of the present study.

Table 7 shows examples of gait parameter estimation. Stroke patients exhibit shorter stride length, step length, and lower cadence compared to healthy individuals, which is consistent with known characteristics of impaired gait. This indicates that the model preserves both the absolute values of gait parameters and the differences between patient and control groups. To further illustrate the capability of *mmGait* in capturing gait characteristics, Table 8 presents step-level stride length variations across consecutive walking cycles. Stroke patients show noticeable fluctuations in stride length across steps, reflecting asymmetric gait patterns, while healthy individuals exhibit more consistent stride behavior. This demonstrates that *mmGait* can capture step-level gait dynamics beyond average statistics.

As in the gait-speed analysis, Table 9 reports the RMSE and ICC values for stride length, step length, and cadence. The RMSE values are generally consistent with the median errors presented in Fig. 10(a) and (b), indicating that the overall estimation errors remain small across walking tests. Stride length demonstrates good agreement with the reference system, with RMSE values below 6 cm and ICC values of 0.885 for patients and 0.793 for healthy controls. Cadence also shows good agreement for healthy controls (RMSE = 4.57 steps/min, ICC = 0.860), whereas the agreement decreases for stroke patients (RMSE = 12.63 steps/min, ICC = 0.469), which is consistent with the occasional step-counting errors observed in patient trials. In contrast, step length

Table 9

Agreement analysis between gait parameters estimated by *mmGait* and the OptiTrack reference. Root mean square error (RMSE) was calculated as the square root of the mean squared difference between the predicted and reference values. Intraclass correlation coefficients (ICCs) were calculated using a two-way random-effects model for absolute agreement of single measurements. RMSE is reported in the original measurement units, and ICC values are presented with their 95% confidence intervals.

Group	Parameter	RMSE	ICC(95% CI)
Patients	Gait speed (m/s)	0.029	0.992 (0.98–1.00)
Controls	Gait speed (m/s)	0.071	0.931 (0.83–0.97)
Patients	Stride length (cm)	5.7	0.885 (0.78–0.94)
Controls	Stride length (cm)	4.7	0.793 (0.54–0.92)
Patients	Step length (cm)	7.6	0.235 (–0.10–0.57)
Controls	Step length (cm)	10.7	0.143 (–0.09–0.47)
Patients	Cadence (steps/min)	12.628	0.469 (0.16–0.70)
Controls	Cadence (steps/min)	4.570	0.860 (0.66–0.95)

exhibits relatively weak agreement despite its moderate estimation error (RMSE = 7.6 cm for patients and 10.7 cm for healthy controls). The ICC values are only 0.235 for patients and 0.143 for healthy controls, with the corresponding 95% confidence intervals including negative lower bounds. These results suggest that the proposed method estimates stride length and cadence more consistently than step length, and that further improvement is needed for accurate step-length estimation. The comparatively low step-length agreement arises from the definition of step length interacting with the sensing characteristics of the system. Step length is computed as the Euclidean distance between the left and right foot positions at a single inter-foot-distance peak, and therefore depends on the accurate *absolute* inter-foot geometry at one instant. In contrast, stride length and gait speed are relative and temporal measures derived from the displacement of the *same* foot over time, in which systematic per-frame localization offsets largely cancel. Because *mmGait* reconstructs temporally smooth foot trajectories, the two predicted feet are drawn toward a common central location, which compresses the peak inter-foot separation. This produces both a systematic underestimation of step length (predicted values are smaller than the ground truth in 94% of walking tests, with a mean bias of –6.4 cm for patients and –9.2 cm for controls) and a narrowed prediction range (predicted step-length standard deviation of 3.2 cm versus 5.0 cm in the ground truth for patients). Since the ICC for absolute agreement penalizes both a systematic bias and a reduced between-subject variance, these two effects jointly explain the low step-length ICC, whereas stride length and gait speed remain accurate under the same underlying localization error.

Overall, these results demonstrate that *mmGait* can accurately estimate clinically relevant gait parameters from the predicted 3D foot trajectories. The low median estimation errors, representative examples, and agreement analysis consistently show that the proposed system preserves both the overall gait characteristics of stroke patients and healthy controls and the step-to-step temporal variations within individual

walking trials. Among the evaluated parameters, gait speed and stride length exhibit the strongest agreement with the OptiTrack reference, whereas cadence and particularly step length remain more sensitive to irregular gait patterns and step-event estimation.

3.5. Computational latency

The computational latency of the current implementation consists of two components: data preprocessing and model inference. To characterize the processing time, we measured the latency of each component on our experimental platforms. Data preprocessing, including FFT operations and dynamic RAmMap generation, was performed on a workstation equipped with an Intel Xeon E5-1620 v4 CPU (3.5 GHz, 32 GB RAM), resulting in an average processing time of 203.8 ms per sample. Model inference was evaluated on a server equipped with an NVIDIA RTX A6000 GPU, where the average inference time was 12.8 ms per sample, corresponding to an inference throughput of approximately 78 samples/s.

Under the evaluated hardware configuration, data preprocessing accounts for most of the measured processing time, whereas model inference contributes a smaller portion of the computational latency. The proposed system is intended for continuous gait assessment, where gait parameters are estimated from complete walking segments to monitor rehabilitation progress instead of providing immediate feedback. Therefore, the measured computational latency is adequate for the intended clinical application.

3.6. Preliminary stroke status classification

In real-world scenarios, including both home-based and clinical environments, multiple individuals such as caregivers or family members may appear within the sensing range of the mmWave device. However, continuous gait monitoring is primarily intended for patients. Therefore, we conducted a preliminary stroke-status classification study to evaluate whether gait characteristics measured by mmWave sensing can distinguish stroke patients from healthy individuals. Stroke patients typically exhibit distinct gait characteristics, such as reduced walking speed, shorter step length, and asymmetric foot placement, compared to healthy individuals. These differences enable the use of a lightweight neural network to learn discriminative gait patterns for classification. We trained this model using the collected dataset consisting of 9 stroke patients and 9 healthy controls. Each input sample covers a 4 s segment of mmWave data to capture longer temporal gait patterns for stroke-status classification. Because the raw mmWave stream is collected at 20 Hz, each segment contains 80 raw frames, which are downsampled to 10 frames before being input to the classification network.

For evaluation, we randomly select 2 stroke patients and 2 healthy controls as the test set, while the remaining subjects are used for training. This process is repeated 50 times with different random splits to assess robustness. Fig. 11 presents the box plots of classification performance across these runs. The central line represents the median,

and the box spans the interquartile range (Q1–Q3). Whiskers extend to values within 1.5 times the interquartile range (IQR), while individual points denote results from each split. The median accuracy reaches 83.6%, with consistently high F1-score (91.0% in median) and precision (95.0% in median), indicating balanced classification performance. The recall values demonstrate reliable detection of the stroke-patient class, while the comparatively lower and more variable specificity suggests occasional confusion with healthy individuals. Despite this variability, the relatively compact interquartile ranges across most metrics indicate stable performance under different data splits, supporting the robustness of the proposed model. These results demonstrate the feasibility of distinguishing stroke patients from healthy controls using gait characteristics measured by mmWave sensing.

4. Discussion and limitations

The experimental results demonstrate the feasibility of contactless gait assessment using mmWave sensing for stroke rehabilitation. Although the proposed system exhibits moderate errors in 3D foot localization, the predicted foot trajectories preserve the temporal walking patterns and inter-foot relationships required for estimating clinically relevant gait parameters. Consequently, *mmGait* achieves reliable estimation of gait speed and stride length, while providing acceptable performance for cadence estimation, particularly for healthy controls. These results suggest that the proposed system is well suited for continuous monitoring of coarse spatiotemporal gait characteristics, such as gait speed, stride length, and rehabilitation trends, which are widely used to evaluate functional recovery after stroke. However, the relatively low agreement observed for step length indicates that the current system is less suitable for applications requiring accurate step-level biomechanical analysis. Despite these encouraging results, several limitations of the current study should be considered.

Comparison with prior systems. The proposed system differs from earlier RF- and mmWave-based gait sensing in both its sensing target and its clinical population. Wi-Fi-based systems such as WiGait [2] estimate walking velocity and stride length from whole-body reflections but do not resolve individual feet, which limits step-level and symmetry analyses, and their coarser spatial resolution is less reliable for the slow, impaired gait (< 0.4 m/s) typical of stroke survivors. mmWave mesh-reconstruction methods such as mmMesh [26] recover full-body pose but rely on the SMPL body model, which is fitted to healthy anatomy and localizes the feet only coarsely because of specularities and limited vertical resolution. Gait-based mmWave identification such as MU-ID [27] targets user discrimination rather than metric gait-parameter estimation. In contrast, *mmGait* performs foot-level 3D localization, is validated on a genuine stroke cohort with asymmetric and temporally irregular gait, and is benchmarked against a sub-millimeter OptiTrack reference. A direct numerical comparison across these systems is complicated by differences in sensing targets, reference systems, and study populations, so a standardized benchmark for contactless gait assessment remains an open direction.

Temporal resolution. The current implementation acquires mmWave reflections at 20 Hz, which is sufficient for estimating the coarse spatiotemporal gait parameters investigated in this work, including gait speed, stride length, and cadence. However, the current sampling rate limits the estimation of more fine-grained gait biomarkers that require higher temporal resolution. Since the employed mmWave hardware supports substantially higher acquisition rates, increasing the sampling frequency is expected to improve the estimation of these gait biomarkers without changing the proposed contactless sensing framework.

Foot representation. The proposed framework predicts a single 3D location for each foot, defined as the center of the corresponding OptiTrack marker cluster. Although this representation is sufficient for recovering foot trajectories and estimating coarse spatiotemporal gait parameters, it does not provide sufficient spatial information for

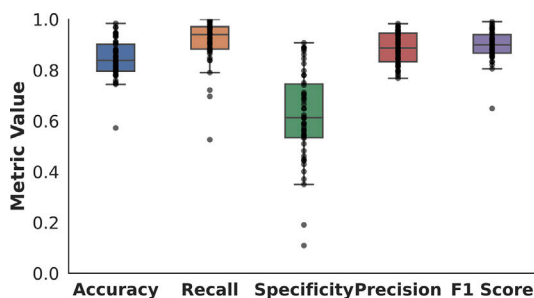


Fig. 11. Box plots of preliminary stroke-status classification performance across 50 random splits of the dataset.

estimating more detailed gait biomarkers, such as heel strike, toe-off, and foot orientation. This limitation is also related to the current OptiTrack marker configuration, where only a small number of markers were attached to each foot to obtain the marker-cluster center as the prediction target. A denser marker configuration during data collection could provide richer ground-truth information, enabling the prediction of multiple anatomical landmarks or the full foot pose for more comprehensive gait analysis.

Scope of applicable gait biomarkers. The two factors above jointly determine which clinical gait biomarkers the current system can and cannot estimate. Gait speed, stride length, and cadence are directly supported and validated against OptiTrack, and step-level asymmetry and gait variability can be characterized qualitatively from the step-to-step inter-foot-distance variations (e.g., Table 8). Base of support (step width) is in principle recoverable from the lateral coordinates of the two predicted foot positions, but it was not validated in this study because a single marker-cluster center was used per foot rather than a medial/lateral landmark. In contrast, fine temporal events such as heel strike, toe-off, and the associated stance/swing timing cannot be resolved with the present 20 Hz, single-point foot representation, and their estimation would require both higher-rate acquisition and richer, multi-landmark foot ground truth.

Dataset size and diversity. The proposed framework was evaluated using data collected from nine stroke patients with varying levels of functional impairment and nine healthy controls. Furthermore, after quality control, one healthy control did not contribute any valid processed samples because reliable OptiTrack foot trajectories could not be obtained, and only synchronized samples with reliable ground-truth foot locations were retained for training and evaluation. Consequently, the amount of usable data remained limited, which may affect the generalization of the proposed model to broader patient populations. A larger and more diverse dataset covering different ages, stroke severities, walking abilities, and rehabilitation stages will be important for further validating the robustness of the proposed framework.

Stroke-status classification. The stroke-status classification experiment should be regarded as a preliminary feasibility study. The current model distinguishes stroke patients from healthy controls based on gait characteristics, demonstrating that mmWave-derived gait features contain discriminative information for stroke-status classification. However, this setting differs from practical deployment, where multiple individuals, including caregivers or family members, may appear within the sensing range and some may also exhibit abnormal gait patterns. Distinguishing the intended patient from other individuals therefore remains an open challenge, requiring multi-person sensing, individual-level recognition, and more diverse subject populations. In addition, the healthy controls in this study were substantially younger than the stroke patients (mean age 28.4 versus 62.6 years; Tables 1 and 2), so the discriminative signal learned by the classifier may partly reflect age-related rather than stroke-specific gait changes. An age-matched control group is therefore needed to attribute the classification performance specifically to stroke.

5. Conclusion

This paper presents *mmGait*, a contactless mmWave-based system for continuous gait monitoring through 3D foot localization. By leveraging mmWave sensing and a deep learning architecture (*mmGaitNet*), our approach enables accurate estimation of clinically relevant gait parameters without requiring wearable sensors or camera-based systems. Experimental results on data collected from stroke patients and healthy controls demonstrate that *mmGait* achieves reliable 3D foot localization under realistic sensing conditions. While absolute localization errors remain moderate, the system effectively captures temporal gait dynamics and inter-foot relationships, enabling estimation of key gait parameters, including gait speed, stride length, step length, and cadence. Agreement analysis demonstrates excellent agreement with

the OptiTrack reference for gait speed and good agreement for stride length, supporting the applicability of *mmGait* for coarse spatiotemporal gait assessment. Importantly, the model preserves both the temporal structure and magnitude of gait patterns, allowing it to capture pathological asymmetry in stroke patients as well as natural variability in healthy individuals. These results highlight the potential of *mmGait* as a practical and non-invasive tool for at-home gait assessment and rehabilitation monitoring. Future work will investigate higher temporal resolution of data acquisition, richer foot representations, and more diverse clinical datasets to further improve detailed gait analysis and model generalization.

CRedit authorship contribution statement

Zhuangzhuang Gu: Software, Investigation, Formal analysis. **Ryan Titus:** Methodology. **Hem Regmi:** Software, Investigation. **Souvik Sen:** Supervision, Project administration, Investigation. **Sanjib Sur:** Supervision, Project administration, Methodology, Investigation.

Ethics statement

All procedures performed in this study involving human participants were conducted in compliance with relevant laws and institutional guidelines and in accordance with the ethical standards of the institutional research committee. The study was reviewed and approved by the University of South Carolina Institutional Review Board (Protocol No. Pro00133989), with ethical approval granted on 20 February 2024. The study was conducted in collaboration with Prisma Health Hospital. The reporting of this work follows the recommendations of the International Committee of Medical Journal Editors (ICMJE). Informed consent was obtained from all individual participants included in the study prior to data collection. The privacy and confidentiality rights of all human subjects were observed throughout data collection, analysis, and reporting, and no individually identifying information is included in the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work is supported in part by the NSF awards CAREER-2144505, MRI-2018966, and NeTS 2342833.

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